

OpenSees Script Review

3-Storey Masonry Frame — *newcode3storey.tcl*

Foreword

The structure of this script is solid and the modelling approach is correct — Macroelement3d for piers and spandrels, three analysis phases (gravity → settlement → pushover), displacement control with a Newton fallback. The issues below are the kind of subtle traps that catch experienced OpenSees users too, mostly because the framework is flexible enough to accept commands silently even when they have no effect. Nothing here reflects a gap in understanding of the structural problem — it is purely OpenSees bookkeeping.

The note on version and extensions matters: two of the items below may or may not apply depending on the exact Macroelement3d implementation in use, and those are flagged explicitly.

Critical Issues

These three items will either produce silently wrong results or cause the pushover to run past its intended target.

1 — Foundation Settlement Is Never Applied

Lines 80–81 and 258

```
fix 1 1 1 1 1 1 1 ;# node 1: ALL six DOFs locked, including vertical (DOF 3)
...
pattern Plain 20 Linear {
    sp 1 3 -0.001 ;# tries to impose a vertical displacement on node 1
}
```

With the **Transformation** constraint handler (used throughout the script), fixed DOFs are eliminated from the system of equations before the solver runs. The **sp** command inside the pattern therefore has nothing to act on — the settlement is accepted without error but never actually applied.

The fix is to leave the vertical DOF of node 1 free:

```
fix 1 1 1 0 1 1 1 ;# DOF 3 (Z, vertical) is now free for settlement
fix 2 1 1 1 1 1 1 ;# node 2 remains fully fixed
```

2 — Floor Load Formula Produces Values ~12× Too Large

Line 218

```
set floorLoad [expr -5.0*$g*$rho*$L_span*$T_pier]
```

Working through the units in SI (N, m, kg):

$\rho \times g$	→	N/m ³	(specific weight of masonry)
$\times L_{\text{span}} \times T_{\text{pier}}$	→	N/m	(load per metre of height)
$\times 5.0$	→	N	(only if 5.0 is a height in metres)

The formula computes the weight of a masonry block **5 metres tall**, with cross-section $L_{\text{span}} \times T_{\text{pier}}$. Since $H_{\text{story}} = 3.0$ m, this is larger than one full storey applied at every floor node — approximately **44 100 N per node**. The structure is therefore pre-loaded far beyond its gravitational state before the

pushover even begins.

If the intent was a **5 kN/m² floor load** (a common value), the correct line is:

```
set floorLoad [expr -5.0e3 * $L_span * $T_pier]    ;# → -3 750 N per node
```

If the intent was the **self-weight of the masonry wall** above each floor (tributary height = H_story):

```
set floorLoad [expr -$rho * $g * $H_story * $L_span * $T_pier]
;# → -26 487 N per node
```

3 — Pushover Fallback Overshoots the Target Displacement

Lines 316–326

```
set nSteps [expr int($targetDisp/$incr)]    ;# = 2000 steps

set ok [analyze $nSteps]                    ;# runs up to 2000 steps, fails at step k

if {$ok != 0} {
    algorithm Newton -initial
    set ok [analyze $nSteps]                ;# runs ANOTHER 2000 steps from step k
}
```

If the first block fails after, say, 1 500 of 2 000 steps, the controlled node is already at 0.15 m. The fallback then requests 2 000 more steps, potentially driving the roof to 0.35 m — well past the 0.20 m target. The output file will contain a trajectory that extends far beyond the intended pushover range.

A step-by-step loop is more reliable:

```
set completedSteps 0

while {$completedSteps < $nSteps} {

    set ok [analyze 1]

    if {$ok == 0} {
        incr completedSteps
    } else {
        puts "Newton failed at step $completedSteps – trying Newton -initial"
        test NormUnbalance 100.0 200 5
        algorithm Newton -initial
        set ok [analyze 1]

        if {$ok == 0} {
            incr completedSteps
            algorithm Newton
            test NormUnbalance 100.0 50 5
        } else {
            puts "Analysis stopped at step $completedSteps / $nSteps"
            break
        }
    }
}

puts "Pushover completed: $completedSteps of $nSteps steps."
```

Significant Issues

4 — Possible Double-Counting of Self-Weight

Line 223 — version-dependent

```
element Macroelement3d 1 ... -density $rho -cmass
...
eleLoad -ele $e -type -selfWeight 0.0 0.0 [expr -$g]
```

The **-density** flag tells the element its mass density (used to build the consistent mass matrix via **-cmass**). The **eleLoad -selfWeight** separately applies a body-force acceleration to each element. In some Macroelement3d implementations, -density is used only for the mass matrix, so eleLoad -selfWeight is the correct way to apply static gravity. In others, the element automatically generates gravity loads from its density, making the eleLoad redundant and the self-weight double-counted.

Action: check the documentation of the specific Macroelement3d version in use. If gravity reactions are roughly twice the expected value after the gravity phase, this is the cause.

5 — Internal Spandrel Nodes Receive No Floor Load

Lines 226–233

```
load 3 0.0 0.0 $floorLoad 0.0 0.0 0.0
load 4 0.0 0.0 $floorLoad 0.0 0.0 0.0
;# nodes 15, 16, 17 (spandrel midpoints) receive nothing
```

Nodes 15, 16, and 17 are the midpoint internal nodes of the three spandrel elements. If -density and -cmass are active, these nodes carry mass. The eleLoad -selfWeight distributes gravity body forces to them through the consistent mass matrix, but any additional floor loads (live load, slab weight not included in self-weight) are missing at these nodes. If the floor loads at nodes 3–8 represent superimposed loads from slabs or occupancy, a proportional share should go to nodes 15–17 as well.

6 — Convergence Tolerance May Be Too Tight for Nonlinear Analysis

Lines 240 and 296

```
test NormUnbalance 100.0 50 5
```

A tolerance of 100 N on the unbalanced force norm is an absolute criterion. With axial pier capacities on the order of $f_c \times A = 6 \times 10^6 \times (1.0 \times 0.25) = 1.5$ MN, this is extremely strict (0.007% of capacity). Near yield or after cracking, the Newton-Raphson iterations may struggle to reach this tolerance within 50 iterations, causing premature divergence even when the structural solution is physically well-behaved. Consider switching to a relative criterion:

```
test RelativeNormUnbalance 1.0e-4 100 5
```

Minor Points

7 — Shear Modulus Implies an Unusually High Poisson's Ratio

Lines 24–26

```
set E 1700.0e6 ;# 1 700 MPa
set G 550.0e6 ;# 550 MPa
```

From the elastic relation $G = E / (2(1+\nu))$, the implied Poisson's ratio is $\nu = E/(2G) - 1 = 1700 / (2 \times 550) - 1 \approx 0.55$. Typical masonry values are $\nu \approx 0.15$ – 0.20 , which for $E = 1$ 700 MPa gives $G \approx 739$ MPa. A value of $G = 550$ MPa is sometimes used as a cracked shear stiffness (~ 0.3 – $0.4 \times G_{\text{elastic}}$), which is a legitimate modelling choice — but it is worth confirming this is intentional and consistent with the reference source for these material properties.

8 — Fracture Energy $G_c = 6 \text{ J/m}^2$ Is on the Low End

Line 29

```
set Gc 6.0
```

Mode-II fracture energy for masonry in the Macroelement3d/TREMURI formulation is typically in the range 20–100 J/m². A value of 6 J/m² will produce a very brittle post-peak response — a sharp, steep drop on the pushover curve immediately after peak force. This is not necessarily wrong, but if the curve shows an unrealistically abrupt capacity loss, G_c is the first parameter to revisit.

9 — No Element Recorders Defined

Lines 200–211

Only nodal displacement and base reaction are recorded. Without element recorders it is not possible to know which piers or spandrels have cracked or yielded, or where in the structure damage concentrates. Adding at least the following will make it much easier to interpret the pushover curve and diagnose unusual behaviour:

```
recorder Element -file PierForces.out -time -ele 1 2 3 4 5 6 basicForce
recorder Element -file SpanForces.out -time -ele 7 8 9 basicForce
```

Summary Table

#	Lines	Severity	Issue	Suggested fix
1	80, 258	Critical	Fixed node: settlement never applied	Release DOF 3 on node 1
2	218	Critical	Floor load ~12× too large (unit mismatch)	Use $5.0e3 \times L_{\text{span}} \times T_{\text{pier}}$
3	316–326	Critical	Fallback reruns full step count, overshoots target	Replace with step-by-step loop
4	223	Significant	Possible double self-weight (version-dependent)	Check Macroelement3d docs
5	226–233	Significant	Spandrel midpoint nodes have no floor load	Add loads at nodes 15, 16, 17
6	240, 296	Significant	Absolute tolerance may be too tight post-cracking	Switch to RelativeNormUnbalance
7	24–26	Minor	G/E implies $\nu \approx 0.55$ vs. typical 0.15–0.20	Verify source of material data
8	29	Minor	$G_c = 6 \text{ J/m}^2 \rightarrow$ very brittle post-peak	Check experimental reference
9	200–211	Minor	No element recorders	Add recorder Element for piers/spandrels

Issues 1, 2, and 3 are the most likely explanation for incorrect or unexpected results. Fixing these three will substantially change the output and should be the first priority before revisiting material parameters or convergence settings.